

TRACE THE ACE

Transcript Intelligence Challenge

TRACE THE ACE BASELINE REPORT

From Dataset Preparation to Model Diagnosis

Dataset | Features | Baseline Model | Error Analysis | Next Steps

Team SkyMap

Trace the Ace Competition

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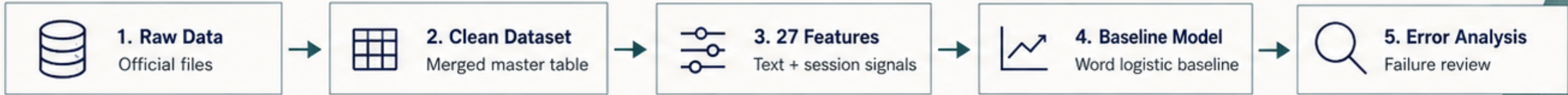
What We Built and Where We Stand

A simple baseline pipeline for predicting whether a student demonstrated a learning objective during a tutoring session.

	Prediction task We predict whether a student demonstrated a specific learning objective during a tutoring session.
	Model input Learning objective + tagged tutor-student transcript + 27 engineered features.
	Model output A probability that the response is correct for that objective.
	Why a baseline? We first built a clean, reproducible system to understand the dataset, measure performance, and identify where the task is difficult.

Current Baseline Snapshot	
Best baseline model	Word Logistic
Local validation log loss	0.5455
Public leaderboard log loss	0.6163
Public AUROC	0.6006
Current rank	144
Lower log loss is better.	
Top-15 target: below ~0.6050 log loss	
Our current baseline is useful, but it still needs a measurable improvement to become competitive.	

Baseline journey



	Project statement	We built a baseline model to estimate whether a student demonstrated a specific learning objective during a tutoring session.
	Key takeaway	The baseline learned useful language patterns, but it struggled to separate different learning objectives within the same session.

Dataset Integration and Feature System

How the official files were combined into a clean response-level master dataset, and how 27 engineered features were built from the tutoring transcripts.

35,072

Response Rows

22,821

Sessions

396

Learning Objectives

70.25%

Positive Labels

Label distribution

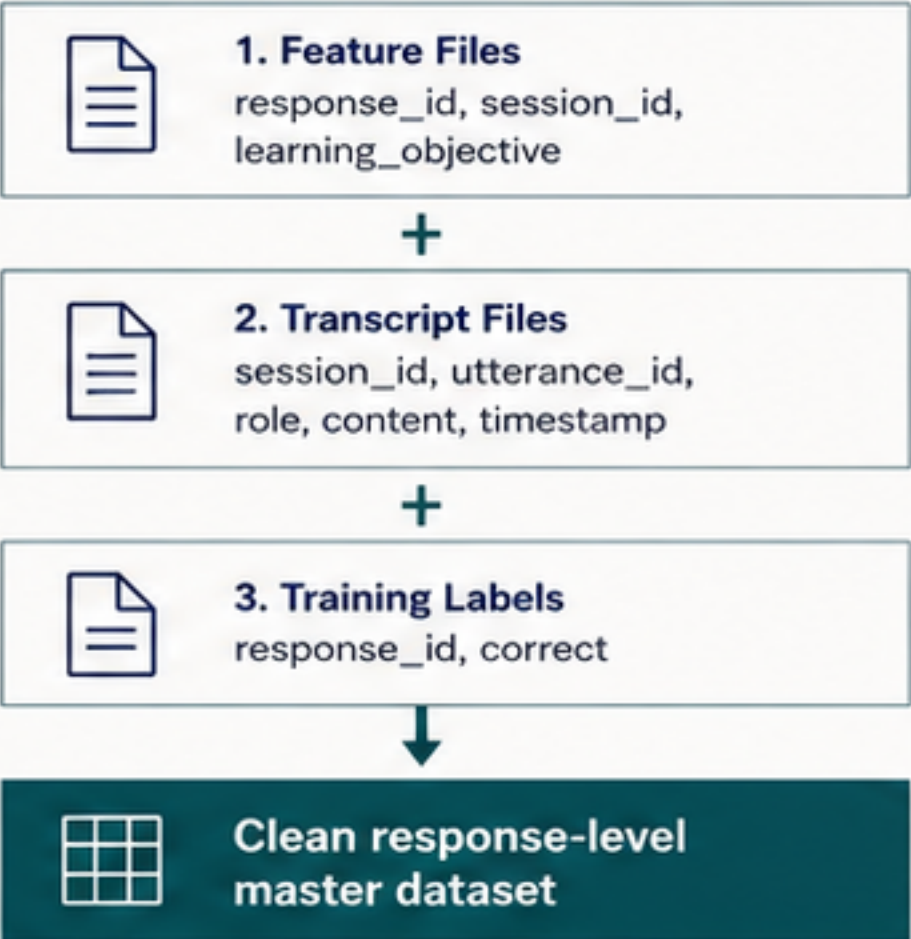
70.25%

29.75%

Positive 70.25%Negative 29.75%

One tutoring session can support multiple response rows.

Official inputs



We used only the official competition files: feature files, transcript files, and training labels.

How we prepared the master dataset

	1. Rebuild each session transcript Transcript files were loaded and sorted by utterance order.
	2. Clean and tag the conversation Speaker roles were standardized and the transcript was rebuilt with tutor and student tags.
	3. Build session-level transcript features We created numeric features from session size, participation, response style, question flow, dialogue dynamics, and transcript quality.
	4. Create the full session table Each session was stored with its full tagged transcript, student-only text, tutor-only text, 20 numeric features, and 7 grouped feature bands.
	5. Merge with response-level training data Training features and labels were merged by response ID, then linked to session-level transcript features by session ID.

What one final row contains

#	response_id
🔗	session_id
🎯	learning_objective
✓	correct label
🗨️	full tagged transcript
👤	student-only text
👤	tutor-only text
📊	20 numeric transcript features
📦	7 categorical feature bands
One final row represents one assessment response, linked to the full tutoring context of its session.	

The 27 Engineered Features

We built 27 interpretable features from the tutoring transcripts and grouped them into 7 families.

1. Session Size and Pace (4) How large and how fast the session is.	2. Student and Tutor Participation (6) How much each speaker contributes.	3. Student Response Style (4) How the student typically answers.	4. Tutor Questioning and Response Flow (2) How often the tutor asks questions and how the student responds.	5. Conversation Dynamics (3) How the dialogue moves between tutor and student.	6. Transcript Quality (1) How much of the transcript is background rather than direct dialogue.	7. Threshold-Based Feature Bands (7) Grouped versions of key numeric features for easier modelling.
Total = 27 features						



Important validation rule

The same session was never placed in both training and validation data. This prevented the model from seeing the same conversation during both training and evaluation.

What EDA Revealed

Key patterns, structural complexity, and early clues from the training dataset.



1. Positive labels were more common

The dataset is label-imbalanced: 24,637 responses are positive and 10,435 are negative. Because about 70% of the labels are positive, a baseline model can naturally drift toward positive predictions if it is not careful.



2. Many sessions were linked to more than one objective

8,364 sessions were multi-objective, which is 36.65% of all sessions. Among them, 3,207 sessions were mixed-label, meaning the same conversation supported both correct and incorrect objectives. This makes the task harder than standard text classification.



3. Transcript length varied widely

The dataset includes short, medium, long, and very long conversations. The response counts were 8,729, 8,972, 8,755, and 8,616 respectively, so no single transcript-length group dominated the dataset. A longer conversation did not automatically make prediction easier.



4. Transcript text was the strongest signal

Conversation words and short phrases carried the clearest predictive signal. The structured-feature baseline reached a log loss of 0.5924, while the word-based baseline reached 0.5455. This showed that text features were more useful than structured features alone.



5. Many numerical features were session-level

A large part of the engineered feature set describes the whole tutoring session. Because those values repeat across multiple objective rows from the same session, they do not easily separate one objective from another. This is one reason why objective-specific discrimination remained difficult.

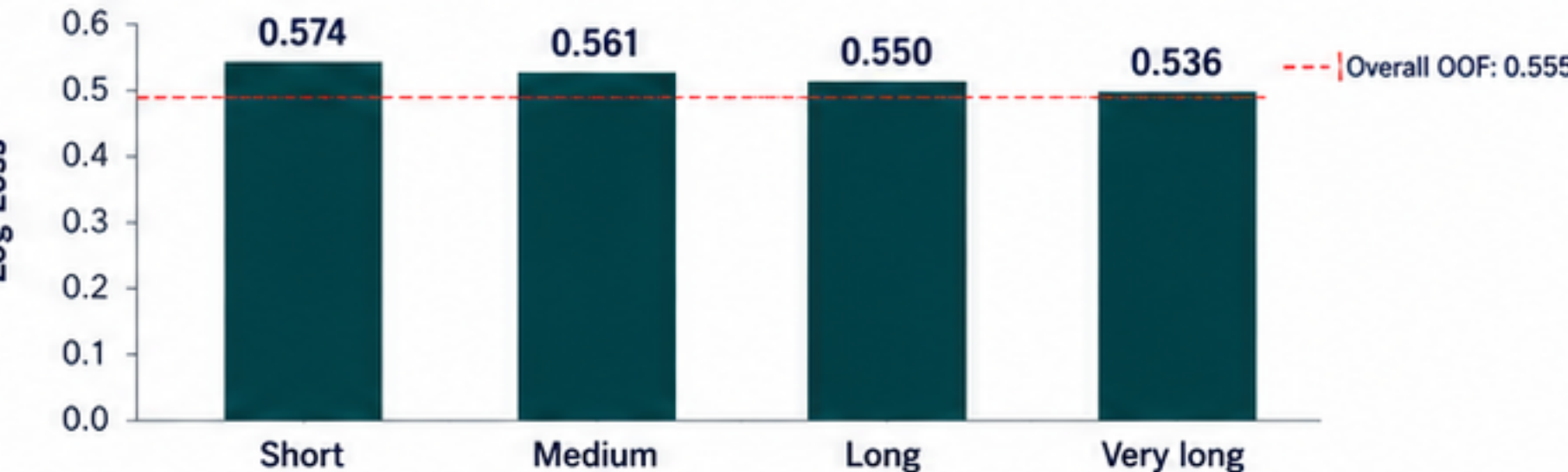
Key EDA facts

 35,072 total responses	 22,821 unique sessions	 396 learning objectives	 14,457 single-objective sessions	 8,364 multi-objective sessions	 3,207 mixed-label sessions
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Label distribution



Transcript length groups



Overall log loss improved slightly for longer transcripts, but negative-class loss remained high in very long sessions (1.141).

Why this mattered for modelling

EDA showed that transcript text was informative, but the task structure was complex. The same conversation could support multiple objectives, and many engineered features described the full session rather than the specific objective.



Main takeaway

EDA showed that transcript text was informative, but the dataset structure was more complex than a standard classification task.

EDA

EXPLORATORY
DATA ANALYSIS

04

From Tutoring Text to Model-Ready Features

How the learning objective, tutoring transcript, and structured session information were converted into reproducible model inputs...

 Total input rows 35,072	 Training rows 28,125	 Validation rows 6,947	 Training sessions 18,256	 Validation sessions 4,565
Training positive rate: 69.95%			Validation positive rate: 71.46%	

1 Build objective-aware text

The learning objective is placed with the tutoring text so the model can evaluate the conversation in context.

 **Main text**
[OBJECTIVE]
learning objective
[TRANSCRIPT]
full tagged transcript

 **Student view**
[OBJECTIVE]
learning objective
[STUDENT]
student-only text

 **Tutor view**
[OBJECTIVE]
learning objective
[TUTOR]
tutor-only text

 Empty training model text: 0

 Empty validation model text: 0

 Avg training document length: ~3,864 words

2 Prepare structured session features

Basic structured view

- 20 original numerical features
- Median replacement for missing values
- Standard scaling
- 6 categorical bands
- One-hot encoding

Output size: **38** features

Engineered structured view

- 20 original numerical features
- Extreme values clipped at training-set 1st and 99th percentiles
- 5 additional log-transformed features
- Median replacement for missing values
- Robust scaling
- 6 categorical bands

Output size: **43** features

38 = 20 numeric + (6 bands × 3 levels)

43 = (20 numeric + 5 log) + (6 bands × 3 levels)

3 Rebuild categorical bands from training data

The original session_turn_volume_band was removed.
The remaining 6 bands were rebuilt using training data only.

 Session duration

 Student short-turn behaviour

 Student turn share

 Tutor questioning

 Student response length

 Dialogue switching

Threshold rule (train-derived)

LOW	MEDIUM	HIGH
bottom third	middle third	top third

 Validation data used the thresholds learned from training data. It did not create its own thresholds.

4 Convert text into numerical signals

Aa Word TF-IDF

- Words, short phrases, and single-character tokens
- Unigrams and bigrams
- Min doc freq: 2 • Max vocab: 60,000
- Sublinear term frequency

 Train: 28,125 × 60,000
Valid: 6,947 × 60,000

 TF-IDF gives more weight to useful words or text patterns and less weight to patterns that appear in almost every document.

{...} Character TF-IDF

- Spelling variation and partial words
- Short mathematical expressions and punctuation patterns
- Char n-grams (3–5)
- Min doc freq: 3 • Max vocab: 60,000

 Train: 28,125 × 60,000
Valid: 6,947 × 60,000

Validation guardrails

 **Session-based split**

Training and validation were separated using session_id, not individual response rows.

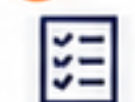
 **No shared conversations**

Session overlap between training and validation was exactly 0.

 **Train-only fitting**

All preprocessing was fitted using training data only:

- Clipping limits
- Medians & scaling values
- Category thresholds
- One-hot categories
- Word & character vocabularies

 **Row consistency checks**

All output matrices matched the saved labels exactly.

- Train rows: 28,125
- Validation rows: 6,947

 **Later five-fold confirmation**

After the initial 80/20 split, the selected baseline was checked with five session-grouped folds. Each session appeared in validation once across that evaluation.

 Often called out-of-fold evaluation.

Four model-ready views were produced

 **Structured Basic**
28,125 × 38

 **Structured Engineered**
28,125 × 43

Aa **Word TF-IDF**
28,125 × 60,000

{...} **Character TF-IDF**
28,125 × 60,000

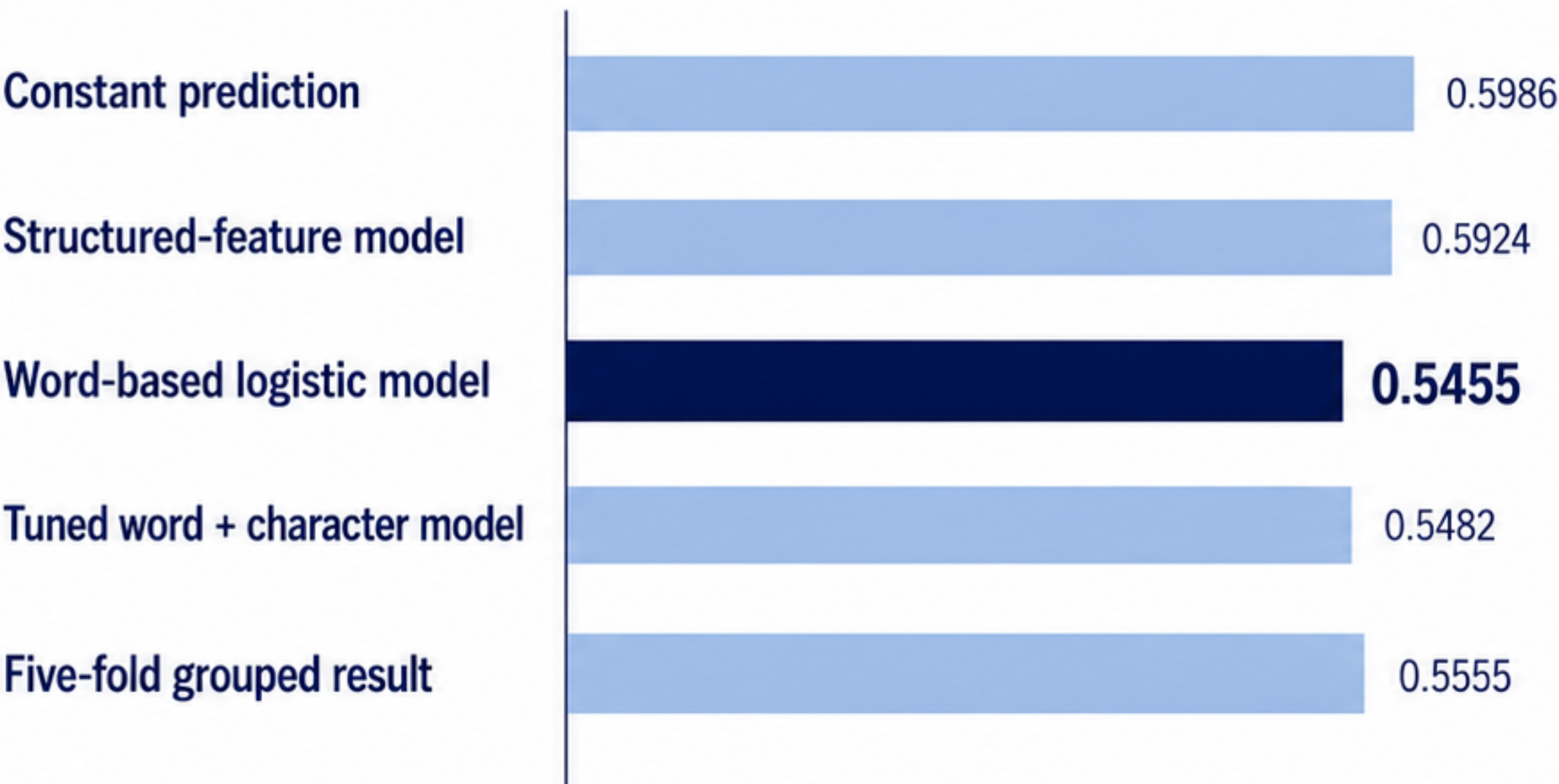
 Feature engineering produced several comparable model inputs while keeping every preprocessing decision separate from the validation sessions.

Baseline Model and Performance

A simple comparison of the baseline candidates and the final model we retained.

Model comparison by Log Loss

Lower is better



Best holdout Log Loss:

0.5455

Grouped five-fold Log Loss:

0.5555

Best model family:

Text-based logistic

Decision:

Retain original baseline

What the metrics mean

Log Loss: measures accuracy and confidence. Lower is better.

AUROC: measures how well the model separates positive and negative cases. Higher is better.

Main findings

- The word-based logistic model became the official baseline.
- Structured features alone were weaker.
- Adding word + character tuning did not improve the baseline.
- The original word-based baseline was retained.

Key takeaway

Text-based modelling gave the strongest starting point, but the tuned variant did not beat the simpler word-based baseline.